

Education

Bachelor of Science in Mechanical Engineering, Taylor University

Anticipated Graduation: May 2027

Minors in Mathematics, Physics, and Honors

GPA: 3.90/4.00, Dean's List

Professional Experience

- **Mechanical Engineering Intern, Quantinuum** **May 2026-August 2026**
 - Worked as a mechanical design engineer, supporting photonics testing/R&D and electronics packaging at leading quantum computing company
 - Owned design of mechanical solutions by integrating optical, electrical, and pneumatic components to increase test cadence by 10X and enable advanced photonics testing
 - Reduced system downtime, increased testing capabilities, and safeguarded crucial electronic components through design of mechanical components for electronics test setups and commercial system deployment
 - Designed cryogenic test rig for thermal strap qualification and characterization of thermal conductance as a function of bolt torque
 - Owned development of UHV test rig to verify components for future cryogenic and vacuum systems
- **Undergraduate Researcher, Engineering Department, Taylor University** **February 2026-Present**
 - Developing thermal management solutions for an orbital robotics system
 - Owning development of research group's first comprehensive, system-wide thermal model in MATLAB
 - Performing design, analysis, and TVAC testing of an actively actuated thermal louver for deployment in low-earth-orbit
 - Researching and defining environmental, thermal, vacuum, and reliability requirements for robotic arm design sub-teams
- **Mechanical Engineering Intern, Danner** **May 2025-August 2025**
 - Owned mechanical design for mechanical integration of GNC/autonomy hardware and drivetrain subsystem components for off-road utility vehicles at electric vehicle startup
 - Developed an in-house mobile battery charger to save manufacturing team ~6 hours a week, participated in its fabrication, and troubleshoot assembly issues
 - Researched and selected parts and suppliers for future vehicle iterations
 - Created Jira workflow to manage engineering change requests, reducing lead time for engineering department by ~8 hours per week

Student Activities & Leadership

- **Hovercraft Competition** **February 2025-May 2025**
 - Responsible for design of thrust and lift subsystem components, increasing hovercraft speed by 35-50%, and leading to 2nd place finish in competition
 - Collaborated with electrical team members to develop hardware integration and wire harness routing solutions
 - Led engineering analysis to optimize selection of motor/propellor combination and inform system design
 - Troubleshoot electrical and mechanical manufacturing and assembly issues
- **Teaching Assistant** **August 2024-Present**
 - Assisting professors with grading and facilitation/set up of labs
 - Leading review/teaching sessions as needed
 - Courses include University/General Physics I & II, Intro to Engineering, Statics, and Modern Physics
- **Honors Guild Leadership Cabinet** **September 2023-Present**
 - Leading coordination of academic events for University facility and guest speakers
 - Served as a peer mentor and teaching assistant for study abroad trip in Italy (J-Term 2025)
 - Sharing information and resources with prospective Honors Students

Technical Skills

- *Mechanical Design:* OnShape, Fusion 360, SolidWorks, AutoCAD, GD&T, PLM (Arena)
- *Simulation & Analysis:* Ansys Fluent, Autodesk FEA/CFD, EES, Hand Cals
- *Programming:* MATLAB/Simulink, C/C++, Python, Mathematica, Excel, Jira
- *Manufacturing & DFM:* 3D Printing, CNC Machining, Welding, Laser/Plasma Cutting, Waterjet, Sheet Metal